

Auteur: Tiffany Kegels
 Studierichting: Informatica
 Oefeningenreeks 1D nr. 19.5

$$z^3 - z^2 - z - 2$$

$$\begin{array}{c|cccc} & 1 & -1 & -1 & -2 \\ 2 & & 2 & 2 & 2 \\ \hline & 1 & 1 & 1 & 0 \end{array}$$

$$z_1 = 2$$

$$z^2 + z + 1 = 0$$

$$D = 1^2(-4) \cdot 1 \cdot 1 = -3$$

$$z_2 = \frac{-1 + i\sqrt{3}}{2}$$

$$z_3 = \frac{-1 - i\sqrt{3}}{2}$$

$$\Rightarrow (z - 2)(z^2 + z + 1)$$

References